GENERATIVE AI & THE LEVEL DESIGNER: AN EXPLORATORY REPORT OF TOOLS, WORKFLOWS, AND CHALLENGES

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Ahad Bandeali

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ABSTRACT

Generative AI promises to democratize game development, yet a critical gap remains for novice designers. This study investigates this “Novice Gap” via a scoping review and an autoethnography. Findings reveal that rather than acting as a “co-creative colleague,” current AI tools function as volatile “slot machines,” producing broken 3D assets that necessitate a “Frankenstein process” of manual repair. This shifts the novice's role from author to technician. Additionally, reliance on centralized AI infrastructure introduces systemic risks of dependency for the industry. To navigate these technical and structural challenges, this thesis proposes a framework of “Curation, Constraint and Repair,” concluding that until automated middleware matures, the democratization narrative remains premature.

Keywords: generative AI, level design, autoethnography, game development, novice workflow.

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TABLE OF CONTENTS

LITERATURE REVIEW..... 1

Introduction..... 1

The traditional level design pipeline..... 2

The “Expert” Promise: GenAI as a “Co-Creative Colleague”..... 4

The “Novice Gap”: Practical, Creative, and Ethical Controversies..... 5

The Practical & Technical Gap..... 5

The Creative & Ethical Gap..... 7

The Infrastructure Gap..... 8

Conclusion and justification for Methodology..... 9

METHODOLOGY..... 11

Data Generation..... 13

Data Analysis..... 13

AUTOETHNOGRAPHY..... 15

Technical Challenges..... 15

Creative Challenges..... 18

Workflow Gaps..... 21

FRAMEWORK..... 25

Rule 1: Curation..... 25

Rule 2: Constraint..... 26

Rule 3: Frankenstein Workflow..... 26

DISCUSSION..... 28

CONCLUSION..... 30

Roadmap for future work..... 31

Final Thoughts..... 32

REFERENCES..... 33

List of Figures

Figure 1 ..... **16**

Figure 2 ..... **17**

Figure 3 ..... **20**

Figure 4 ..... **21**

Figure 5 ..... **22**

LITERATURE REVIEW

Introduction

The intersection of artificial intelligence (AI) and game design has evolved substantially over the past decade, moving from the basic non-player character (NPC) behaviours toward sophisticated generative models that produce rich and adaptive content. Traditionally, game development is an interdisciplinary effort that requires a wide range of specialists, including artists, designers, and engineers, to collaborate on a complex, resource-intensive process (Karlsson et al., 2022; Korsah, 2025; Alharthi, 2025). This complexity is especially true in 3D level design, where the "authoring burden" - the immense, time-consuming manual effort required to create vast, believable, and unique game worlds - is one of the medium's core challenges (Johynson-bey et al, 2023).

Generative AI (GenAI) has emerged as a revolutionary force with the potential to solve this exact problem. Modern GenAI, powered by large language models (LLMs) and diffusion, promises a new paradigm. The new "expert" promise is that GenAI can act not just as a tool, but as a "creative partner" (Alharthi, 2025) or "co-creative colleague" (Larsson et al., 2022), which acts as an active participant in the design process rather than a passive tool. However, much of the literature mentioned does not state the potential limitations of such technology on a broader scale. Brennan et al. (2025) argue that AI is constituted by the logics of a few dominant tech firms and that the introduction of these systems concentrates power among the deployers of the tech, leaving users (in this case, game studios/ developers) insecure, vulnerable and unable to contest the determinations made by the "smart machine". This goes against the optimistic view set by Alharthi (2025) and Larsson et al. (2022) that AI is a colleague and a partner, when in fact it is technology built to be relied on.

The autoethnographic findings of this study indicate that treating the AI as a partner often masks a reality of dependency on an uncontrollable system- a dependency that manifests in unforeseen technical hurdles rather than creative liberation.

Furthermore, a significant gap exists between this “expert” promise and the practical reality for designers. The academic literature is currently fragmented. On one hand, it is highly focused on the technical implementation of isolated (Korsah, 2025) expert systems. On the other hand, a separate body of work discusses the high-level, abstract ethical controversies surrounding AI, such as bias, trust, and originality (Tsamados et al., 2022)

What is missing is what Alharthi (2025) calls a designer-centred perspective that bridges these two conversations. There is a lack of research that holistically investigates the entire modern creative pipeline - from 2D concept to 3D asset to in-engine implementation - from the practical viewpoint of a novice or independent designer.

This literature review will synthesize the existing research to build a clear picture of this gap. First, it will establish a baseline by defining the traditional, manual 3D level design pipeline. Second, it will explore the “expert” literature, outlining new GenAI-assisted workflows and tools, and the “co-creative” promise. Finally, it will analyze the document's practical, creative, and ethical challenges of this new paradigm, identifying the “Novice Gap” that this thesis aims to address.

The traditional level design pipeline

To understand the disruptive potential of generative AI, one must first define the traditional level design workflow it seeks to replace. The conventional 3D game content pipeline is a multi-stage, a highly technical process that has been the industry standard for decades (Karlsson et al., 2022; Labschutz et al., 2011).

As documented by Labschutz et al. (2011) in their case study of the game Dynamite Pete, this level design pipeline involves a sequential progression of specialized tasks. However,

it is essential to recognize that these pipelines vary from studio to studio and game to game. The traditional level design pipeline begins with a Concept Phase, where a team draws outlines of the plot, setting, and mechanics into Concept Art, which artists create sketches and coloured reference images to guide 3D modellers. The 3D asset creation process follows a specific sequence: Low-Polygon Modelling builds the basic geometry; UV Layout unwraps the model into a 2D map; High-Polygon Modelling creates normal maps to achieve incredible detail; and Texturing involves painting materials onto the UV Layout. These assets are then utilized in White-Boxing for layout testing and Set-dressing for final decoration. The process is finalized with lighting (in this case, via global illumination) and implementation, during which all models and textures are tested in gameplay. However, it is essential to note that technology, especially real-time technology, continues to evolve and streamline this pipeline as game engines develop.

This traditional level design process is defined by its challenges. It is immensely resource-intensive, with the Dynamite Pete case study requiring 1550 person-hours for a single level (Labchutz et al., 2011). Furthermore, Karlsson et al. (2022) note that it is “vulnerable to suboptimal interdisciplinary communication”. Studios often struggle to combine design, art, audio, and narrative elements. Ersoy (2024) adds that these high costs usually force developers to outsource asset creation, which can lead to style inconsistencies and lower quality, creating a strong economic driver for automated solutions. This is where Generative AI comes into play, promising to fill that gap. However, Goenka (2024) discusses this ongoing “hype vs reality” discussion against Generative AI and notes that while asset creation is the “holy grail” of the gaming infrastructure, Generative AI’s lack of quality is making game studios hesitant. He also notes that studios are wary of relying on tools that cannot deliver the unique, highly detailed fictional worlds they need, reinforcing the risk of dependency on average outputs. This hesitation aligns with the initial empirical observations of this study, where the AI’s output often appeared generic or average compared to the bespoke quality required for professional game development.

The “Expert” Promise: GenAI as a “Co-Creative Colleague”

The generative AI paradigm proposes a radical shift in the creative process. The literature suggests that GenAI tools act as a ‘democratizing force” by lowering technical barriers (Alharthi, 2025; Begemann & Hutson, 2024) and empowering solo developers and small teams to create high-quality games with fewer resources.

This new role for AI is defined in the literature as Mixed Initiative Co-Creativity (MI-CC), which Larsson et al. (2022) describe as a joint effort by a human user and AI to create content together. In this model, the AI is elevated from a simple tool to a co-creation partner or colleague. Culturally, this promise is framed as a shift in the nature of creativity itself. Kelly (2022) argues that these tools “open up possibilities for more art, not less,” envisioning a future where a user can simply type a prompt to “make a 1970s kitchen…in full volumetric detail.” He frames the workflow not as manual labour, but as summoning art via “magic spells” (prompts), creating a duet between human and machine (Kelly, 2022).

In practice, this “expert” promise has manifested in a new generation of highly specialized tools and fragmented workflows documented across the literature:

Text-to-Level Generation: A dominant theme in the research is the use of LLMs to generate playable game levels directly from natural language prompts. Studies have demonstrated the use of models like MarioGPT and BERT to create 2D platformer levels (Korsah, 2025; Sweetser, 2024) and tools like LLMaker to design levels for dungeon crawlers (Korsah, 2025).

Text-to-Asset Generation: Other tools focus on asset creation, such as using Midjourney and Stable Diffusion to generate 2D concept art, character portraits, and tileable textures from text

Text-to-Code: Models like OpenAI’s Codex, as shown by Ratican and Hutson (2024), are capable of “translating neutral language prompts directly into executable code”. Lee et al. (2023)

suggest a designer could generate playable C# prototypes for Unity 3D without writing code themselves.

Crucially, these tools are not one-stop solutions. The “expert” literature describes complex, multi-step pipelines that chain these “isolated” tools together. One documented workflow for the simulation Neighbourly involves using ChatGPT to generate JSON data from a text prompt, which is then fed into Jinja2 templates to produce final, editable Python code (Johnson-Bet et al., 2023).

Another pipeline, detailed in a case study by Begemann and Hutson (2024), mirrors the traditional workflow: 1) generate 2D concept art in Leonardo AI, 2) convert the art to a 3D model in Luma AI, 3) import the model into Blender for optimization, and 4) integrate the final asset into Unreal Engine 5. The most complex systems, like Google’s Genie, are entirely “expert-facing” and consist of multiple deep-learning models, including a “video tokenizer, a latent action model, and a dynamics model,” to function (Dablander, 2024).

The “Novice Gap”: Practical, Creative, and Ethical Controversies

The promise of a “super game designer” (Lee et al., 2023) who can simply speak a world into existence is undermined by the practical and ethical challenges documented in the same body of literature. This “Novice Gap” can be understood through three interconnected areas of controversy: the practical workflow, the loss of creative control, and the unresolved ethical questions.

The Practical & Technical Gap

The primary gap for a novice designer is that the “expert” workflow is fragmented, technically demanding, and produces unusable assets.

The literature repeatedly notes that GenAI tools are “isolated” and not integrated into a “holistic game development workflow” (Korsah, 2025). Successfully using them requires developers to master complex systems and to possess robust AI development frameworks (Chergarova et al., 2024). This creates a high technical barrier, requiring significant computational resources and possessing power (Mehta, 2025; Chergarova et al., 2024). Ersoy (2024) identifies that compatibility with existing pipelines is a significant barrier, as tools often fail to integrate seamlessly with industry-standard software like Photoshop or game engines. This lack of seamless integration emerged as a recurring theme in the autoethnographic data, particularly when attempting to move assets between platforms where file formats and texture maps frequently broke down.

The most significant documented gap exists between AI-generated assets and game-ready assets. The empirical case study by Begemann and Hutson (2024), which tested the exact 2D-to-3D pipeline, found that the results were a critical limitation. The 3D models from tools like Luma AI were described as “grotesque” and “overly complex geometries”, rendering them impractical for game development without significant modifications. An examination in Blender revealed excessive polygon counts and poor topology, necessitating substantial retopology for any potential game-engine application. Even after manual cleanup, the models failed in-engine, exhibiting significant deformation when animated. The autoethnographic investigation later confirmed this “grotesque” reality, particularly regarding polygon counts that were vastly higher than industry standards (e.g. a 15,000-polygon rock), rendering assets functionally unusable for optimized games. This reflects the “authoring nightmare” described by Johnson-Bey et al. (2023), where an AI generates arbitrary string values that a human developer must then manually create new simulation rules for.

The creative process is not a single, magical command. It is a delicate relationship of tedious trial and error (Larsson et al., 2022). As Kelly (2022) notes, getting an AI to obey specific instructions feels like pushing against a forcefield as the tools favour “surprise” over

“obedience” (Kelly, 2022). This is confirmed by Lee et al. (2023), whose case study of Project Nox required 77 attempts over 6 hours to generate a single usable windmill asset. On the other hand, however, there is a loss of creative control when using these tools. Gómez (2025) argues that users of Generative AI are playing with permutations, hoping to strike a functional application. He states that this “normalizes randomness” and encourages a culture of low-accountability development. This “randomness” became a defining feature of the documented workflow, where iterating prompts functioned less like design and more like gambling for a usable result.

The Creative & Ethical Gap

Beyond the technical failures, the literature identifies a deep-seated controversy regarding the creative and ethical implications of these tools.

The designer’s role shifts fundamentally from authorship to curation (Alharthi, 2025). Alharthi (2025) documents a fear of losing the ‘human touch’ as AI-generated content is often described as generic, derivative, or lacking the emotional depth and artistic nuances of human work.

A central ethical problem is that algorithms are “not ethically neutral” (Tsamados et al., 2022). They are trained on inaccurate, skewed, or systemically biased data that reflects long-lasting structural inequalities. This leads to unfair outcomes and perpetuates biases or generates culturally insensitive content (Moon et al., 2025; Tsamados et al., 2022). Crawford and Paglen (2021) argue that datasets are political interventions built on shaky and skewed assumptions. This structural bias is empirically quantified by Nicoletti and Bass (2023), who found that Stable Diffusion does not just reflect reality but amplifies stereotypes - for instance, generating images of judges that were only 3% female and representing crime-related keywords with disproportionately darker skin tones.

Designers and users are interacting with inscrutable systems (Tsamados et al., 2022). There is a cognitive impossibility for humans to interpret massive algorithmic models, which is an inherent characteristic of self-learning algorithms. This lack of transparency means users cannot know why the AI produced a certain (perhaps biased) result.

Bender et al (2021) provide the definitive critique of this technology, arguing that large language models are “Stochastic parrots” - systems that haphazardly stitch together sequences of linguistic forms without any reference to meaning. This explains the hallucinations and lack of control designers experience; the AI is not an intelligent colleague but a mimic without understanding. This concept of the parrot explains the bizarre hallucinations encountered in this study, such as 3D models with biologically impossible anatomy, confirming that AI lacks any semantic understanding of the objects it creates.

Finally, the literature highlights a profound anxiety about labour and displacement. While most sources frame GenAI as a tool to augment rather than replace the designer (Moon et al., 2025; Sweetser, 2024), Aarthi (2025) documents serious concerns about how automation might reduce opportunities for junior designers and lead to workforce reductions.

The Infrastructure Gap

Beyond the immediate technical flaws of current tools, a more systemic risk emerges from the centralization of AI infrastructure. While the “expert promise” frames AI as a tool for empowerment, critical scholarship argues that the current ecosystem concentrates power in the hands of a few dominant technology firms (Brennan et al., 2025).

This dependency introduces a significant volatility risk into game development pipelines. Unlike traditional software that runs locally and reliably, generative AI relies on complex, cloud-based infrastructure that is prone to catastrophically failing due to outages or policy updates (Ekko System, 2025). The recent strategic partnership between OpenAI and NVIDIA to deploy massive centralized compute infrastructure underscores this trend (NVIDIA Corporation,

2025), while it promises scale. It also funnels the entire industry's creative capability through a single point of failure. For a game studio, building a pipeline on such volatile infrastructure means being at the mercy of external vendors who can alter pricing, deprecate models, or suffer technical setbacks at any moment.

Furthermore, the legal volatility of these tools presents a "rug pull" risk for developers. As noted by Velzen (2024), using AI during development (DevAI) creates a liability where the model might output protected intellectual property- such as a character identical to Mario - exposing the studio to lawsuits long after the game has shipped. This legal uncertainty, combined with the current AI's inability to deliver "unique, highly detailed" assets that studios require (Goenka, 2024), suggests that the "creative colleague" may actually be a liability.

Conclusion and justification for Methodology

A critical analysis of the reviewed literature reveals a fundamental paradox at the heart of the generative AI discourse. While the expert literature heavily promotes the concept of democratization - where AI lowers barriers to entry for non-technical creators (Ratican & Hutson, 2024; Larsson et al., 2022) - the technical literature simultaneously documents a reality of increasing complexity.

My analysis suggests that the "democratization" narrative relies on a significant amount of invisible labour. The literature describes high-level succession (e.g., "AI generated a level") but frequently omits or glosses over the granular, intermediate steps required to achieve them. For instance, while Lee et al. (2023) celebrate the "super game designer," their own data reveals a workflow defined by hundreds of failed iterations. Similarly, Begemann and Hutson (2024) identify the impracticality of AI assets but stop short of defining a new framework for novices to manage this impracticality. Consequently, the literature has created a gap not just in knowledge, but in the process. There is an abundance of theory on what AI can do, but a scarcity of demonstrated, practical reality on how a novice survives the friction of using it.

Therefore, this study argues that the current academic conversation is missing a crucial middle ground. It lacks an empirical account that treats these technical failures not as user errors, but as systemic features of the Novice Gap.

METHODOLOGY

This chapter outlines the methodological framework designed to answer the central research question: *How can a practical framework bridge the gap between the literature on generative AI and the actual workflow challenges and opportunities encountered by a novice 3D Designer?* To investigate this gap, this project employs a two-part qualitative research design that combines a scoping review with an autoethnography. This hybrid approach was chosen first to map the established “expert-oriented” knowledge in the field and then to place it in direct dialogue with the practical, lived experience of a novice.

The exploratory nature of the research question requires a qualitative approach. The goal is to understand the nuances of a creative workflow, the subjective experience of challenge and opportunity, and the complex human-computer interaction at the heart of using generative AI. These phenomena cannot be adequately captured through quantitative metrics alone. This study, therefore, employs two distinct yet contemporary qualitative methodologies.

The first part of this study is a scoping review. A scoping review is a form of knowledge synthesis that addresses an exploratory research question by systemically searching, selecting, and synthesizing existing knowledge (Mak & Thompson, 2022). Its primary purpose is to map key concepts, types of evidence, and research gaps within a defined area or field. For this project, the scoping review systematically maps the “expert-oriented literature” on the tools, workflows, and challenges of using generative AI in 3D level design.

The second part of this study is an autoethnography. Autoethnography is a research method in which the researcher uses their personal experience to understand a broader phenomenon (Wall, 2006). This study employs an analytic approach, which uses the researcher’s experience as data to develop a theoretical or practical understanding of a wider process. It is “analytic” because the researcher’s story is placed in “dialogue with informants

beyond the self" -in this case, the academic literature gathered in the scoping review - intending to produce new analytic insights.

This research is situated within the rapidly evolving digital landscape of generative artificial intelligence for creative practice, specifically within the professional content of videogame development and the use of Unreal Engine 5. The study's location is twofold. The scoping review is conducted within a publicly accessible academic database (e.g., the ACM Digital Library or IEEE Xplore). The analytic autoethnography takes place entirely within the researcher's personal digital workspace, using a specific suite of publicly available software: Stable Diffusion for 2D concept and texture generation, Luma AI for 3D asset generation, Blender for 3D model refinement, and Unreal Engine 5 for final scene assembly.

In this qualitative study, the researcher is the primary instrument for data generation and analysis. The researcher's positionality as an admitted novice in level design is not a limitation but is, in fact, the central strength and core methodological component of this study. This novice perspective enables an authentic investigation of the practical realities, hurdles, and undocumented workflows a new designer encounters when engaging with these powerful AI tools for the first time.

While the researcher is a novice in the specific practice of level design, they possess technical proficiency in the target game engine (Unreal Engine 5) and traditional 3D software (Blender). This positionality is crucial, as it focuses the study squarely on the challenges and opportunities of integrating a generative AI workflow, rather than the difficulties of learning 3D software in general. This allows for a more targeted and nuanced analysis of the "gap".

This study draws from two distinct samples corresponding to its two methodological parts.

For the scoping review, the sample consists of a body of peer-reviewed articles. The selection of these sources is governed by a detailed a priori protocol developed before the review begins. This protocol specifies inclusion and exclusion criteria based on the PCC

(Population, Concept, Context) mnemonic: Population (Level designers, game developers, technical artists), Concept (generative AI tools, workflows, and pipelines for 3D asset and environment creation), and Context (3D videogame development).

For the autoethnography, the sole participant is the researcher. The sample is the body of experimental data generated during the completion of a specific, bounded creative task: the creation of a “small, serene first clearing at dusk” with Unreal Engine 5 using a generative AI workflow.

Data Generation

Data for this study were generated using two distinct methods. First, data were generated through a systematic and comprehensive search of relevant databases. Second, for the autoethnographic content, data were generated through meticulous journal-keeping. This journal documented the entire creative process, systemically capturing the prompts used, the resulting AI outputs, technical encounters, workflow workarounds discovered, and subjective reflections on the experience.

Data Analysis

The data from the literature undergo a thematic synthesis to identify and map the key tools, workflows, and challenges. This synthesis must be more than a superficial summary and should involve analytical interpretation.

The written journal is subjected to a qualitative thematic analysis. The journal entries are systemically coded to identify recurring themes, which are then organized into three key categories: Technical Challenges, Creative Opportunities, and Workflow Gaps.

The final analytical step is a comparative analysis. The themes identified in the autoethnography are placed in direct dialogue with the themes from the scoping review. This

process explicitly identifies and analyzes the “gap” between the expert-oriented literature and the practical novice experience, providing the foundation for the final framework.

AUTOETHNOGRAPHY

This study employed an analytic autoethnography to investigate the practical reality of the democratized generative AI workflow. The researcher, acting as a novice to AI tools but proficient in traditional game engines, undertook a bounded creative task: the creation of a “small, serene forest clearing at dusk” within Unreal Engine 5.

The data generation process adhered to the “expert” pipeline identified in the literature by Begemann & Hutson (2024), which delineates a structured progression from conceptualization to engine integration. The workflow began with Phase 1: 2D Concept Generation, utilizing Stable Diffusion to produce initial visual references. It proceeded to Phase 2: 3D Asset Generation, where Luma AI was employed to convert those concepts into three-dimensional models. Finally, the process concluded with Phase 3: Optimization and Assembly, which involved refining assets in Blender and integrating them into Unreal Engine 5.

The following section presents the thematic analysis of the research journal, documenting the friction points (gaps) where the expert promise of instant creation collided with the novice reality of technical troubleshooting and manual repair.

Technical Challenges

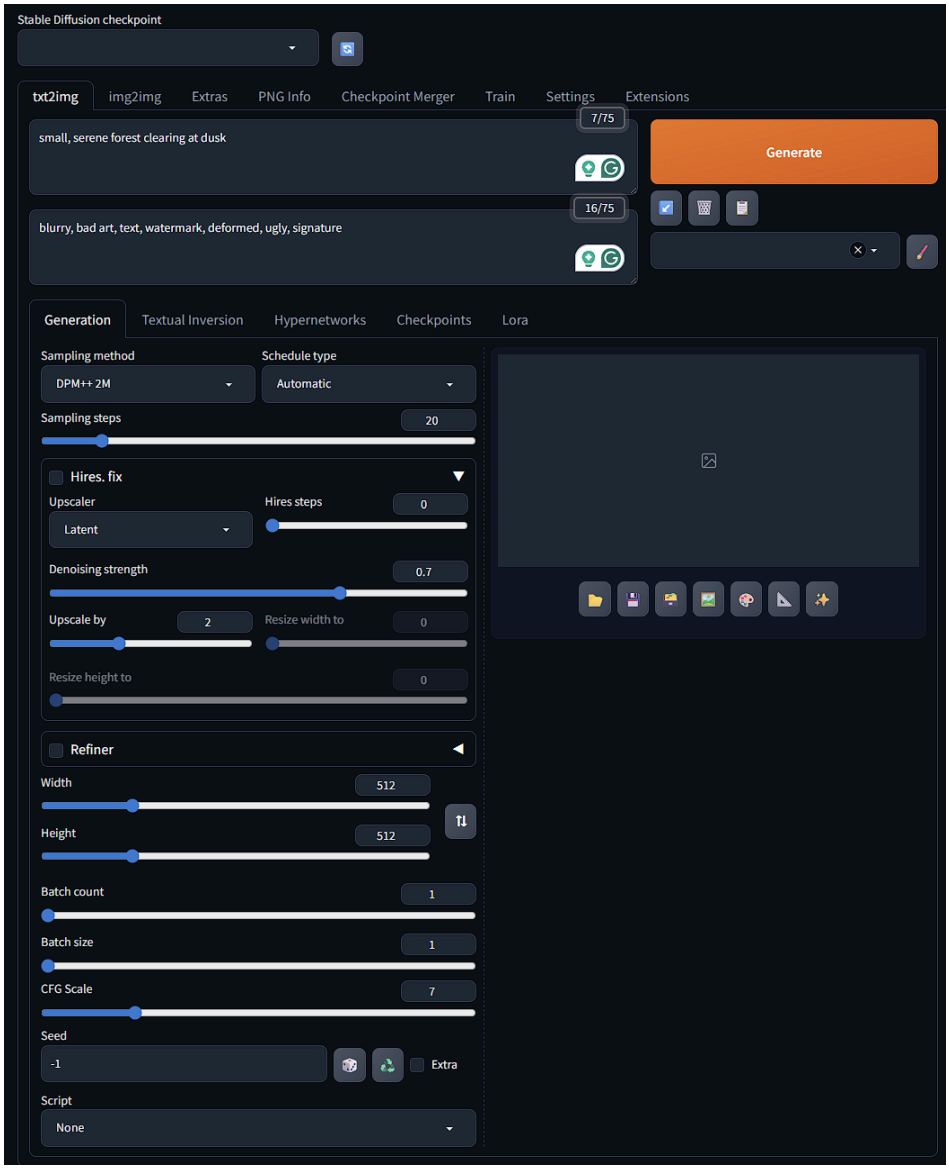
The literature often frames generative AI as a tool that lowers technical barriers for designers (Alharthi, 2025). However, the autoethnographic data reveal that the entry barrier is merely shifted from artistic skill to technical troubleshooting.

The first challenge emerged before the creative process began. Contrary to the promise of accessible software, installing the local instance of Stable Diffusion (Automatic1111) was not “plug-and-play”. The researcher was forced to troubleshoot specific dependencies, discovering that the software required specific legacy versions of Git (2.51.2) and Python (3.10.6) to

function. This confirms the findings of Chergarova et al. 2024, who note that integrating AI requires developers to assess complex “software infrastructure” that is often undocumented in general studies.

Upon installation, the researcher was confronted with what Tsamados et al. (2022) described as inscribable evidence - a UI filled with technical jargon unrelated to art or design. The interface presented 11 distinct technical parameters (under the Image-to-Text category), including “Sampling Method (DPM++ 2M),” “CFG Scale,” and “Hires. Fix,” without explanation (see Figure 1). An attempt to intuitively increase the resolution using the “Upscale by” slider to 4 resulted in a system failure (RuntimeError: Not enough memory), forcing a compromise to a lower resolution (1612x1612)

Figure 1. *The Stable Diffusion (Automatic1111) user interface showing the array of technical parameters*



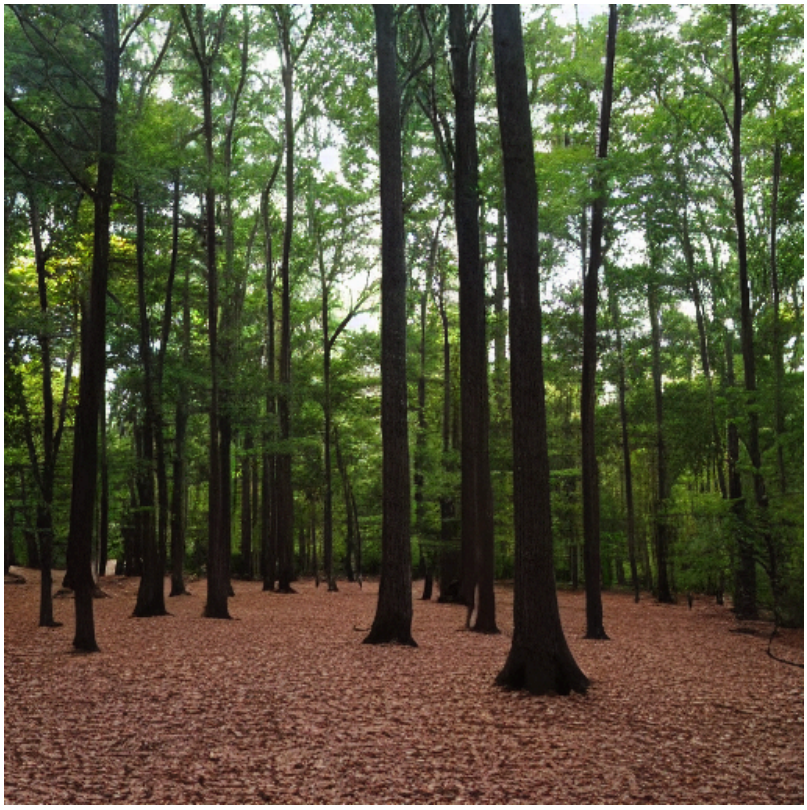
The experience contradicts the “super game designer” narrative (Lee et al., 2023). Instead of a creative partner, the tool acted as a technical gatekeeper, requiring the user to navigate hidden hardware constraints and obscure parameters just to generate a base image

Creative Challenges

Despite the technical friction, the study confirms the “expert” claims that AI can accelerate ideation and facilitate a shift from authorship to curation (Alharthi, 2025)

In Phase 1, the AI successfully demonstrated its value as a brainstorming tool. It took only three iterations to generate a realistic scene that matched the creative vision of a “serene forest”. The third output provided a sophisticated composition with soft lighting and an even spread of vegetation details that would have taken a novice hours to illustrate manually. However, because the process was unpredictable (as shown in Figure 2), the outputs varied drastically in quality.

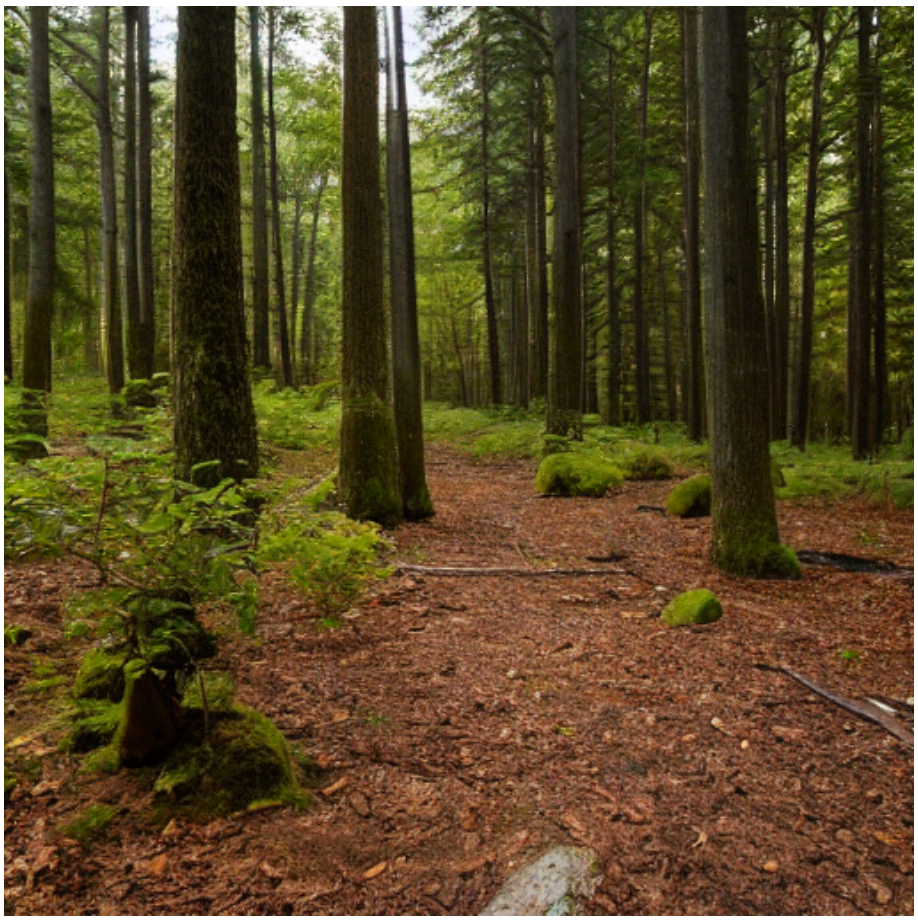
Figure 2. *Three iterations of concept art generated from the same prompt: “small, serene forest clearing at dusk.” generated from Stable Diffusion (AUTOMATIC1111)*



(a) Iteration 1: Lack of vegetation and artificial straightness



(b) Iteration 2: Appearance of unintended artifacts (creature)



Iteration 3: The final selected concept

Similarly, in Phase 2, Luma AI's Genie model allowed for the rapid generation of volume. The researcher was able to create five variations of a mossy tree trunk in approximately 20 minutes per asset. This data supports the efficiency arguments found in the literature review (Korsah, 2025; Ratican & Hutson, 2024), demonstrating that, given the quantity of assets, the AI pipeline is significantly faster than traditional modelling.

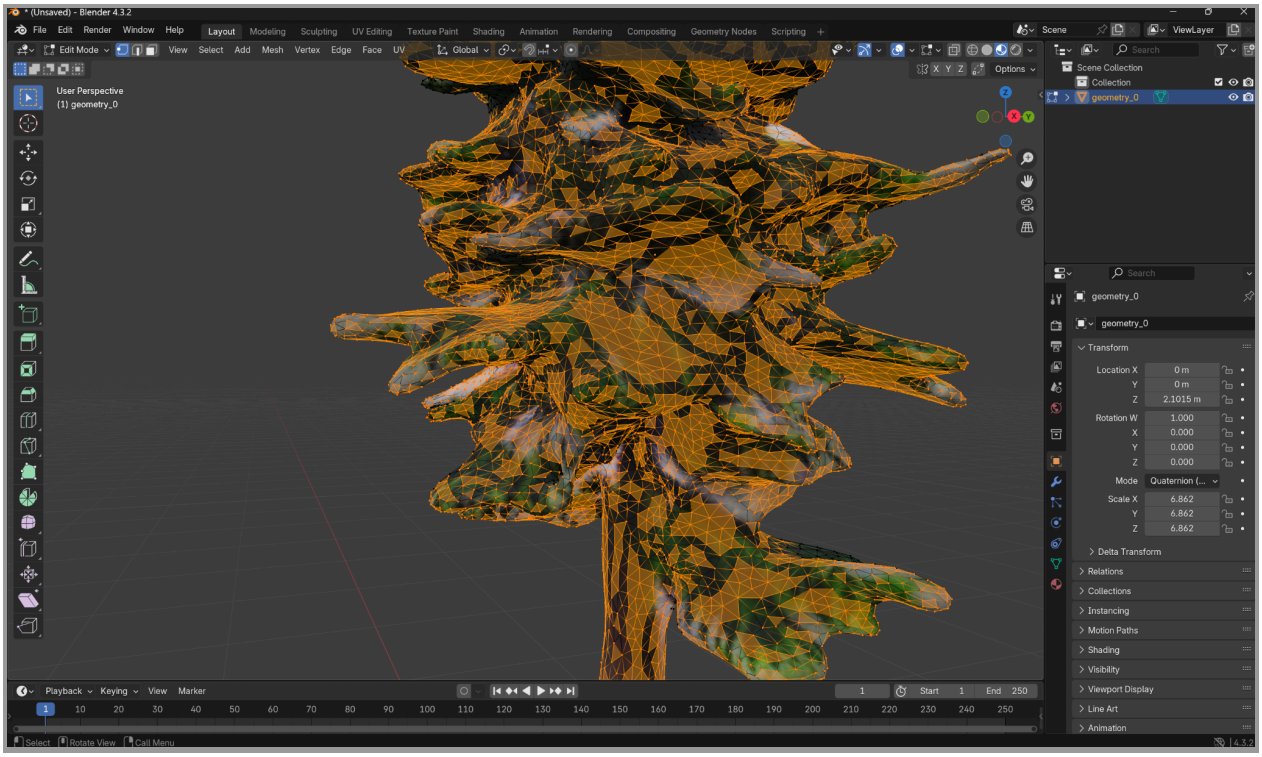
Workflow Gaps

The most significant findings of this study emerged in Phase 3, where the Practical and Workflow gaps identified in the literature (Begemann & Hutson, 2024; Johnson-Bey et al., 2023) were empirically confirmed. The transition from AI generation to game engine implementation was not seamless; it was a “Frankenstein” process of stitching together broken outputs.

The data directly validates Begemann and Hutson’s (2024) finding that AI-generated 3D models are impractical for game development.

While the Luma AI assets appeared acceptable in thumbnails, closer inspection in Blender revealed them to be grotesque and plastic-looking. As illustrated in Figure 3, the wireframe analysis revealed a chaotic structure unusable for standard game workflows. The Western Larch tree (Chase, 2007) prompt failed, defaulting to generic pine trees that looked like deformed sculptures.

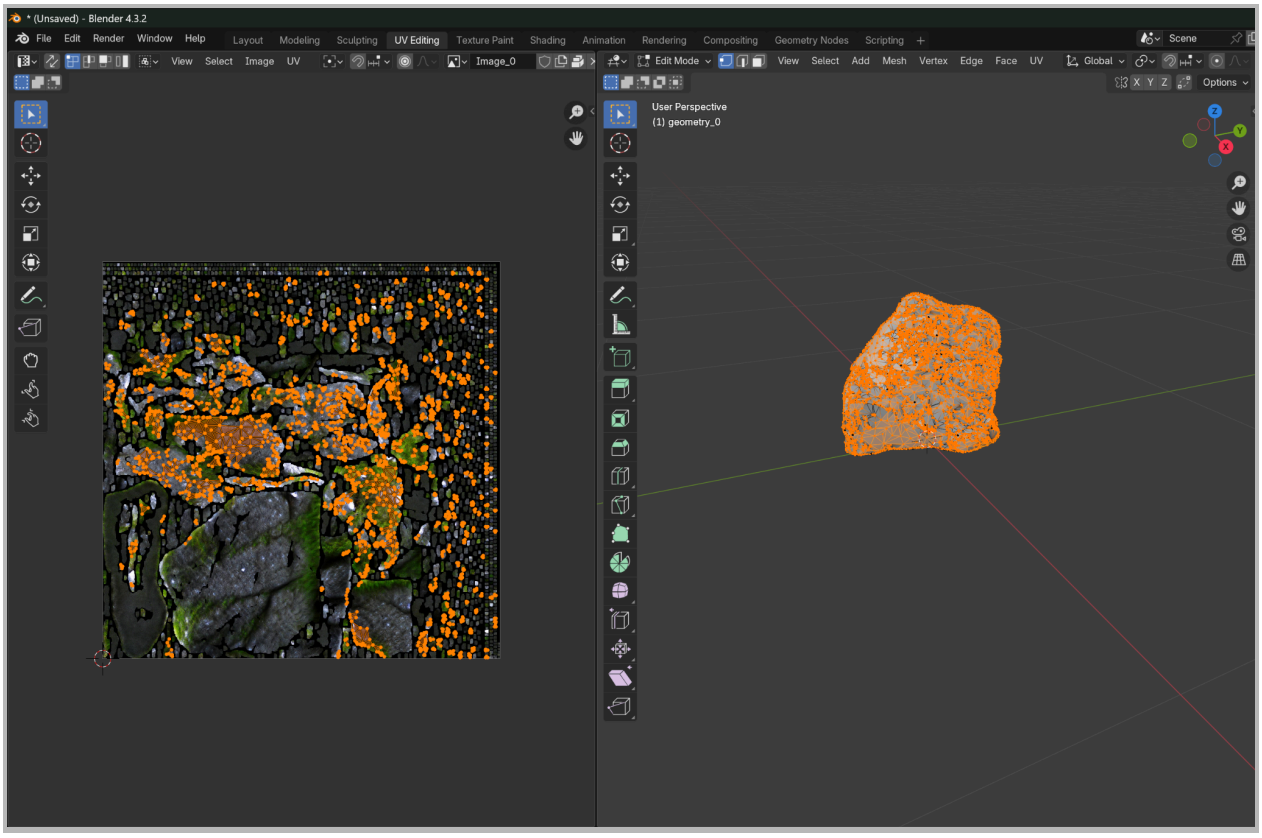
Figure 3. *Wireframe view of the 3D tree model generated by Luma AI in Blender 4.3*



Even after applying automated optimization tools (the Decimate modifier at a ratio of 0.3), a single background rock asset remained at 14,884 faces. In a professional pipeline, such an asset would typically target 100 - 2,000 faces (Labschutz et al., 2011). This confirms that AI lacks context for game-readiness, creating assets that are computationally expensive and unoptimized

This study exposed a critical lack of interoperability between the “isolated” tools (Korsah, 2025). The textures generated by Luma AI failed to import into Unreal Engine (version 5.5.4), forcing the researcher to switch back to Stable Diffusion to generate replacement textures. Furthermore, the UV maps of the AI models were fragmented and chaotic, resembling disconnected islands. This made applying the new textures impossible without visual glitching.

Figure 4. *The UV map of an AI-generated rock asset from Luma AI viewed inside Blender 4.3*

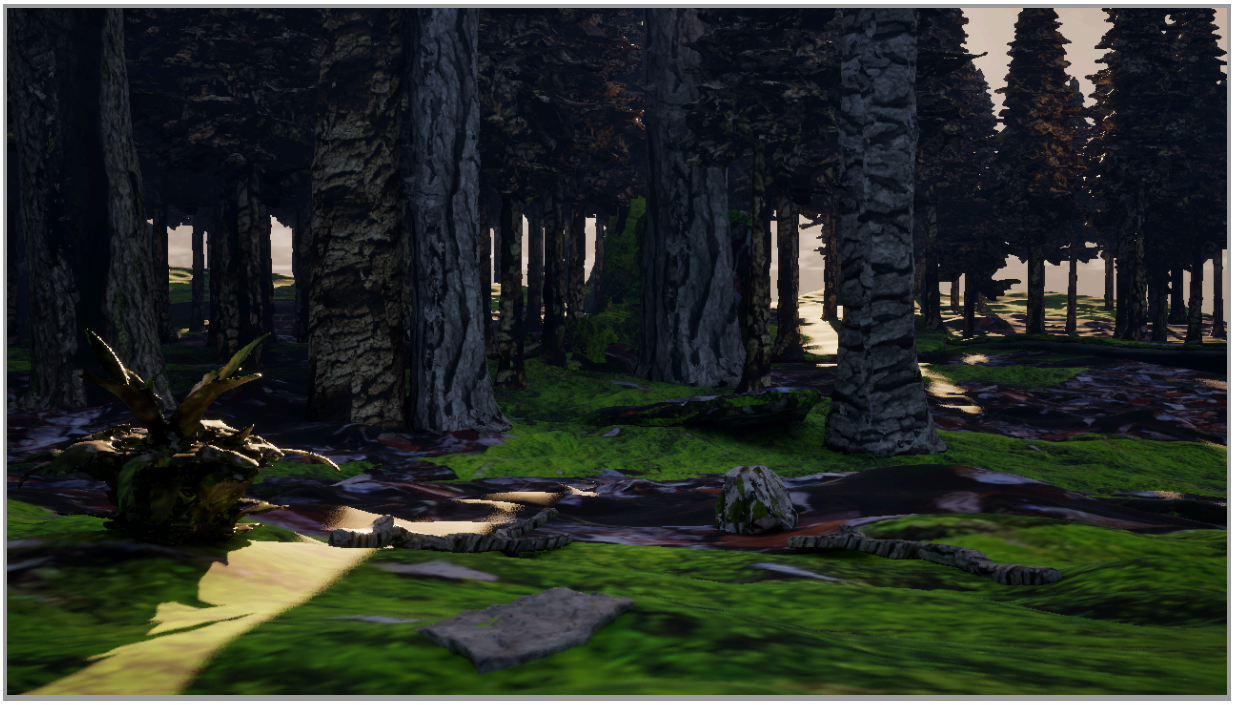


To salvage the assets, the researcher had to override engine settings manually. The ground texture appeared wet and plastic-like due to the lack of a Roughness map, requiring a manual override to 0.9 to simulate natural light interacting with dirt/soil.

Finally, the study encountered the “Stochastic parrot” phenomenon (Bender et al., 2021) firsthand. In Phase 1, the AI spontaneously generated a “deformed creature” with “3 legs” that was not requested in the prompt. This hallucination highlights the model's lack of control and understanding, confirming that the user acts not as a master but as a filter for the system’s random, sometimes disturbing outputs (Nicoletti & Bass, 2023).

The autoethnographic data paints a complex picture. While the AI tools delivered on the promise of speed (generating concepts and models in minutes), they failed on the promise of utility. The final environment, shown in Figure 5, was described in the research journal as “dark, muddy,” and populated by assets that “look like they are floating” or “pasted on.”

Figure 5. *Final renders of the forest scene assembled in Unreal Engine 5*



Render A



Render B

The resulting workflow was not the democratized creation of a “super game designer” (Lee et al., 2023), but rather what this study terms a “Frankenstein process.” This concept builds upon Kelly’s (2022) definition of AI creation as a technological assemblage, but highlights the impractical and “grotesque” nature of the results documented by Begemann and Hutson (2024). As shown in Figure 5, the final artifact is a technological assemblage of incompatible parts manually stitched together.

The researcher was forced to act as a human glue, manually stitching together incompatible file formats (.glb to .fbx), overriding broken material data, and accepting unoptimized, “glitching” assets. This suggests that for a novice, generative AI does not remove the need for technical expertise; it simply demands a new kind of expertise - that of a repair technician for broken AI outputs.

FRAMEWORK

The previous chapters have diagnosed a significant ‘Novice Gap’ in the integration of generative AI into level design. While the literature promises a democratized future where AI acts as a co-creative colleague (Larsson et al., 2022), the practical autoethnography study revealed a reality defined by technical barriers, fragmented workflows and usable assets. The 3D models generated were impractical for game development (Begemann & Hutson, 2024), requiring extensive manual repair that defeated the purpose of automation.

To answer the research question: How can a practical framework bridge the gap between the expert literature on generative AI and the actual workflow challenges and opportunities encountered by a novice-level designer? - This chapter proposes a new framework. This framework transforms the failures documented in the autoethnography into actionable guidelines for novice designers, reframing the role of AI not as a ‘magic wand’, but as a specific, limited tool within a larger manual pipeline.

Based on the empirical findings of this study, the following framework offers three core principles for novice-level designers attempting to integrate generative AI into their workflow.

Rule 1: Curation

Addresses the creative limitations observed in this study, specifically the finding that, while the literature envisions AI as a “super game designer” (Lee et al., 2023), the tool struggles with specific creative intent, such as failing to generate a Western Larch tree. Consequently, the guideline proposes that novices must shift their role from authorship to curation (Alharthi, 2025), abandoning the fight to force the AI to produce a specific vision. Instead, designers should adopt the slot-machine mindset (Doctorow, 2025), treating AI as a generator of volume rather than as pre-recognition. Actionable steps include generating 5-10 variations of an asset immediately and avoiding the “dark art” of prolonged prompt engineering (Lee et al., 2023). Suppose the AI fails to produce a usable concept after constant attempts. In that case, the

designer should pivot the project's aesthetic to match the AI's output, rather than forcing the AI to match the original vision.

Rule 2: Constraint

Addresses the technical limitations confirmed by this study, specifically that AI-generated 3D models suffer from excessive polygon counts and poor topology (Begemann & Hutson, 2024). The creation of the "15k Rock" demonstrated that automated optimization tools, such as Blender's Decemiate modifier, are currently insufficient for producing professional-quality assets. Consequently, the guideline mandates that raw AI-generated models should never be utilized for her assets (objects the player interacts with or views at close range. Instead, action should be taken to restrict AI models exclusively to background debris, such as distinct rocks, rubble and foliage. In these contexts, the grotesque topology and potential shading errors identified by Begemann and Hutson (2024) remain invisible to the player, while traditional modelling or photogrammetry should be reserved for high-fidelity hero assets until AI topology generation improves.

Rule 3: Frankenstein Workflow

Addresses the interoperability gap identified in the literature and confirmed by this study. Research indicates that AI tools remain isolated (Korsah, 2025) and often lack sufficient compatibility with existing pipelines (Ersoy, 2024). The autoethnographic findings corroborated this reality, revealing critical failures such as textures failing to import and UV maps appearing as fragmented, disconnected islands. Consequently, the proposed guideline rejects the fantasy of a one-click solution, instead advocating a Frankenstein Mindset in which the designer serves as the essential human glue connecting incompatible tools. To bridge these gaps, novices must acquire specific repair skills, including manual file conversions (e.g., .glb to .fbx), material overrides to correct missing PBR data (such as manually adjusting roughness setting), and UV

tolerance - learning to utlizise noise-based textures like moss to mask the seams of broken AI-generated UV maps rather than relying in patterned textures that require clean topology.

DISCUSSION

This study's findings offer a critical counter-narrative to the optimism of the literature. While Kelly (2022) envisions a future where users can "summon" worlds with "magical spells," this research demonstrates that the current reality is one of manual labour and technical troubleshooting.

The creative process documented in the autoethnography aligns with the "Slot Machine Metaphor" described by Doctorow (2025). Rather than acting as a deliberate designer, the novice is forced into the role of a gambler, "playing with permutations" of prompts and hoping to strike a functional application (Gómez, 2025). This dynamic normalizes randomness and low accountability, fundamentally altering game development from an engineering discipline into a stochastic wagering process.

The autoethnographic data strongly support the critique by Bender et al. (2021) that large language models are "Stochastic parrots". The spontaneous generation of a three-legged creature in Phase 1 confirms that these systems mimic form without understanding meaning or biology. This "hallucination" phenomenon creates a significant risk for designers, who must vigilantly filter out misguided evidence (Tsamados et al., 2022) and distorted reality (Nicoletti & Bass, 2023).

However, contrary to the fears of "labour displacement" and "workforce reductions" (Alharthi, 2025; Ersoy, 2024), this study suggests that AI does not replace the designer; it creates their workload in new areas. The researcher spent significantly more time fixing AI assets (retopology, texture repair) than would have been required to develop them with traditional tools, such as SpeedTree or Quixel. This suggests that the Novice Gap is not just a learning curve, but a fundamental inefficiency in the current technology.

This Major Research Project set out to investigate how a practical framework could bridge the gap between the expert promises of generative AI and the reality for novice-level

designers. Through a literature review and an autoethnography, the study concludes that the gap is currently bridged only by manual technical expertise.

The “democratization” of game development (Ratican & Hutson, 2024) has not yet arrived in the 3D domain. While 2D concepting is accelerated, the 3D pipeline remains impractical (Begemann & Hutson, 2024) for novices. The proposed framework - focusing on Curation, Constraint and Repair offers a realistic path forward. It encourages novices to embrace AI not as a replacement for skill, but as a chaotic source of raw material that requires a human hand to refine, optimize and integrate into a coherent digital world.

CONCLUSION

This Major Research project began with a central question: How can a practical framework bridge the gap between the expert literature on generative AI and the actual workflow challenges and opportunities encountered by a novice-level designer?

To answer this, this study navigated a complex landscape defined by a sharp divide. On the one hand, the literature presents a narrative of democratization, in which AI serves as a “co-creative colleague” capable of streamlining production (Larsson et al., 2022; Ratican & Hutson, 2024). On the other side, the “novice” reality - documented through an autoethnography- revealed a fragmented ecosystem where technical barriers, “grotesque” 3D topology (Begemann & Hutson, 2024), and ethical opacity (Tsamados et al., 2022) create significant barriers to entry.

The resulting framework (presented in the Framework section) bridges this gap not by addressing AI's technical flaws, but by changing the designer's behaviour. It proposes a shift from authorship to curation and repair, acknowledging that for the novice, AI is currently a source of raw, chaotic material rather than a finished product.

While 2D generative tools have lowered the barrier to entry for concept art, the 3D asset generation pipeline remains impractical for novices without prior technical knowledge. The autoethnographic data confirms that the skill required to fix an AI-generated 3D model (retopolgy, UV unwrapping, shader coding) is often equal to, or greater than, the skill needed to create it manually. Therefore, the claim that AI democratizes 3D game development is, at this stage, a theoretical promise rather than a practical reality.

The unpredictable nature of large language models, demonstrated by the hallucinations encountered in Phase 1 of the study, introduces a new risk layer to the design process. Unlike traditional tools, which do precisely what they are told, AI tools introduce misguided evidence and distorted outputs (Nicoletti & Bass, 2023). This forces the novice designer to adopt a new

role: that of a quality assurance filter who must constantly guard against unwanted artifacts and bias.

Far from being an empowered “super game designer”, the novice user is placed in a position of vulnerability. The reliance on “black box” cloud infrastructure (Ekko System, 2025) and the “slot machine” nature of generation (Doctorow, 2025) mean that the creative process is no longer owned by the creator but leased from a volatile, centralized power structure (Brennan et al., 2025). This shifts the developer’s status from a skilled artisan to a dependent operator of a proprietary machine.

The most significant friction point is not asset generation but its integration. The “Frankenstein” workflow required to move assets between Luma AI, Blender and Unreal Engine confirms that the current ecosystem is deeply fragmented (Korsah, 2025). Until “middleware” solutions (Chergarova et al., 2024) are developed to automate ile conversion and optimization, the Novice Gap will remain a significant bottleneck.

Roadmap for future work

This study was an exploratory investigation into a rapidly evolving field. As such, it has uncovered several significant questions beyond the scope of this project, but critical for future research.

This project addressed the Novice Gap through manual workarounds (the Frankenstein method). A significant question for computer science research is: Can automated tools be developed to bridge the gap reliably? Future research should investigate the development of “bridge” software specifically designed to clean messy AI outputs - automating retopology, auto-rigging, and UV unwrapping - specifically for game engines. This would move the solution from “human repair” to “software repair”.

This autoethnography captured a single snapshot of a novice’s first attempt. It did not address the long-term impact of these tools on learning. A critical question for educators is:

Does the “repair” workflow hinder or help the acquisition of traditional design skills? Future longitudinal studies should track novices over 6-12 months. Does relying on AI “crutches” prevent students from learning fundamental principles of topology and composition? Or does the “curation” role develop a new, equally valuable skill set in critical analysis and art direction?

While this study touched on the ethical risks of bias and ‘misguided evidence’ (Tsamados et al., 2022), it did not propose a comprehensive method for teaching ethical curation. As AI tools become standard, research is urgently needed to develop educational frameworks that teach novice designers how to see and filter bias in AI outputs. Developing a pedagogy of curation - where students are trained not just to generate, but to evaluate synthetic media ethically - is a vital next step for the field.

Final Thoughts

Generative AI represents a paradigm shift in game development, but it is not a magic wand. For the novice designer, it is currently a chaotic, powerful and demanding tool that requires skepticism, technical patience, and a willingness to engage in the messy work of repair. By acknowledging these limitations and adopting the framework proposed in this thesis, new designers can begin to navigate this complex landscape, moving from the frustration of the Novice Gap to a more realistic and productive practice.

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